

Substrate temperature dependent physical properties of sprayed $\text{Zn}_{0.76}\text{Mg}_{0.24}\text{O}$ films

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Received: 9 April 2009 / Accepted: 17 June 2009 / Published online: 3 July 2009
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Abstract The composition, microstructural and optoelectronic properties of $\text{Zn}_{1-x}\text{Mg}_x\text{O}$ thin films grown by spray pyrolysis have been studied. The films were prepared on glass substrates at different substrate temperatures in the range, 200–350 °C for a fixed magnesium composition of $x = 0.24$. The films showed the predominant (002) reflection corresponding to the hexagonal wurtzite structure of ZnO. The preferred orientation doesn't change with the deposition temperature. The films prepared at 300 °C showed good crystallinity with an average surface roughness of 6.2 nm. The optical studies revealed that the optical transmittance increased slightly with the increase of substrate temperature of the films. The variation of energy band gap, photoluminescence and electrical resistivity of the grown layers was also studied.

1 Introduction

$\text{Zn}_{1-x}\text{Mg}_x\text{O}$, a II–VI wide band gap semiconductor, is a potential candidate for the fabrication of optoelectronic devices in the ultraviolet (UV) region of the spectrum. It also possesses the advantages, such as the availability of lattice-matched single crystal substrates [1], tunable band gap energy (3.3–7.8 eV) [2, 3], relatively low thin film

growth temperatures (100–750 °C) [4], which are crucial for practical optoelectronic devices. Moreover, it has good optical properties, i.e. broad and dominated green emission with a weak UV emission and better optical transmittance characteristics in the visible as well as UV region. In general, strong deep-level or green emission in the photoluminescence (PL) spectrum is related to the presence of structural defects and impurities in the corresponding structures. These defects are highly sensitive to the deposition temperature and show strong influence on the optoelectronic properties of the grown layers. To our knowledge, a few reports are available on the preparation of ZnMgO films by chemical spray pyrolysis and their characterization [5]. Hence in the present study, an attempt has been made to prepare $\text{Zn}_{0.76}\text{Mg}_{0.24}\text{O}$ films using a simple and cost effective process, chemical spray pyrolysis at different substrate temperatures. The structural, morphological and optoelectronic behaviour of the grown layers were investigated as a function of substrate temperature and the results were discussed.

2 Experimental

$\text{Zn}_{0.76}\text{Mg}_{0.24}\text{O}$ films were grown by spray pyrolysis technique at different substrate temperatures in the range, 200–350 °C on ultrasonically cleaned glass substrates. 4 N pure $\text{Zn}(\text{CH}_3\text{COO})_2$, $\text{Mg}(\text{CH}_3\text{COO})_2$ and methanol were mixed thoroughly to get the precursor solution with a concentration of 0.1 M and a few drops of HCl were added to the solution to increase the solubility of the compounds. The substrate temperature was maintained using Eurotherm temperature controller with an accuracy of ± 5 °C. The solution and carrier gas flow rates were fixed as 6 ml/min and 8 l/min, respectively. The substrate to source distance

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was fixed as 25 cm. The spray head was moved in the horizontal plane by means of a microprocessor controlled stepper motor system in order to get uniform films on the substrate. The composition of the layers was determined using VG Microtech ESCA2000 X-ray photoelectron spectrometer. The structural and surface topological studies of the films were carried out using Siemens X-ray diffractometer and Veeco atomic force microscope, respectively. The spectral transmittance of the films was recorded using Hitachi UV–Vis–NIR spectrophotometer in order to determine the optical energy band gap, absorption coefficient, refractive index and extinction coefficient. The photoluminescence properties were studied using Fluorolog 3 fluorescence spectrophotometer. The electrical resistivity of the layers was measured using the four-probe method using silver electrodes.

3 Results and discussion

XPS measurements were performed to evaluate the composition and chemical bonding configuration of $\text{Zn}_{0.76}\text{Mg}_{0.24}\text{O}$ films deposited at different substrate temperatures. The binding energies were calibrated by taking the carbon, C 1s peak (284.6 eV) as reference. Figure 1 shows the XPS spectrum of the experimental films deposited at 300 °C recorded in the binding energy range, 0–1,350 eV. The XPS spectrum showed two strong peaks at the binding energies, 1,022 and 1,046 eV that respectively correspond to the Zn 2p_{3/2} and Zn 2p_{1/2} in addition to other normal and Auger reflections. The peak observed at 1,303.5 eV was assigned to Mg 1s reflection and can be attributed to Mg in Mg–O bonding and not to the elemental ‘Mg’. This implies that Mg incorporation in the films occurred at Zn sites. The highest Mg incorporation in ZnO lattice with a doping level of 23.7 at.% was achieved at a substrate temperature of 300 °C. These layers also showed compositional uniformity across the thickness that was observed from sputter etching of the layers for different depth levels by Ar ions.

Figure 2 shows the XRD spectrum of $\text{Zn}_{0.76}\text{Mg}_{0.24}\text{O}$ films grown on glass substrate in the scanning range 30–70°. It can be seen that all these layers showed (002) peak around $2\theta = 34^\circ$ that correspond to the (002) orientation of ZnO hexagonal structure. The effect of substrate temperature that varied in the range, 200–350 °C on the crystalline quality of $\text{Zn}_{0.76}\text{Mg}_{0.24}\text{O}$ films was also studied using XRD analysis, which indicated that the crystallinity of the films depended strongly on the deposition temperature. All the diffraction peaks originated from the hexagonal structure of ZnO with the lattice constant, $a = 5.205 \text{ \AA}$. The layers formed at a temperature of 300 °C were preferably oriented along the (002) plane

although the films showed other peaks that correspond to (100), (101), (102), (110), (103) and (112) orientations of ZnO. The full width at half maximum (FWHM) value calculated for the (002) peak was ~ 0.3 , indicating that the grown layers were highly crystalline. The intensity of the (002) reflection decreased with the increase of substrate temperature. The observed higher intensity of (002) reflection for the layers grown at 300 °C might be due to the enhancement of horizontal mobility of ad-atoms and the condensation coefficient during the cluster formation with the increase of substrate temperature. This leads to an improvement in the crystallinity, which is the usual phenomenon observed in the growth of thin films. For the deposition temperatures higher than 300 °C, a decrease of intensity of the (002) peak was noticed in addition to an increase in the intensity of other peaks. This might be due to the vapourisation of methanol from the precursor solution prior to the sprayed droplet reaching the substrate surface owing to its higher vapour pressure [6], resulting in the formation of the film with powdery nature.

The crystallite size (D) of $\text{Zn}_{0.76}\text{Mg}_{0.24}\text{O}$ films prepared at different substrate temperatures was calculated using the fringe width at half maximum (FWHM) of the (002) peak using Scherrer’s formula [7]. The films prepared at a substrate temperature, $T_s = 300 \text{ }^\circ\text{C}$ showed the highest crystallite size of 45 nm and its value decreased above and below this temperature. The lower grain size at substrate temperatures $< 300 \text{ }^\circ\text{C}$ might be due to the insufficient thermal energy provided to the substrate that caused to an incomplete reaction between the precursor solutions. The surface morphology of $\text{Zn}_{0.76}\text{Mg}_{0.24}\text{O}$ films was analysed using atomic force microscope, which indicated that substrate temperature had a significant influence on the morphology of the grown layers. Figure 3 shows the typical three-dimensional AFM image of $\text{Zn}_{0.76}\text{Mg}_{0.24}\text{O}$ films deposited at 300 °C. The picture shows spherical grains that were uniformly distributed on the substrate. The grain size evaluated from the AFM pictures closely matched with the value calculated from the XRD data. The average surface roughness of the representative film grown at the substrate temperature 300 °C was found to be 6.2 nm.

Figure 4 shows the transmittance spectra of $\text{Zn}_{0.76}\text{Mg}_{0.24}\text{O}$ films, measured in the wavelength between 300 and 2,500 nm, deposited at three different substrate temperatures. The optical transmission falls very sharply near the UV region due to the onset of fundamental absorption. The spectra showed a high transmittance, $> 60\%$ for the films deposited at 300 °C and the films prepared at substrate temperatures $< 300 \text{ }^\circ\text{C}$ exhibited lower transmittance. The higher transmittance observed in the films grown at $T_s \sim 300 \text{ }^\circ\text{C}$ was attributed to lower scattering effects, structural homogeneity and better crystallinity whereas the lower transmittance in the layers

Fig. 1 XPS spectrum of $\text{Zn}_{0.76}\text{Mg}_{0.24}\text{O}$ films deposited at 300 °C

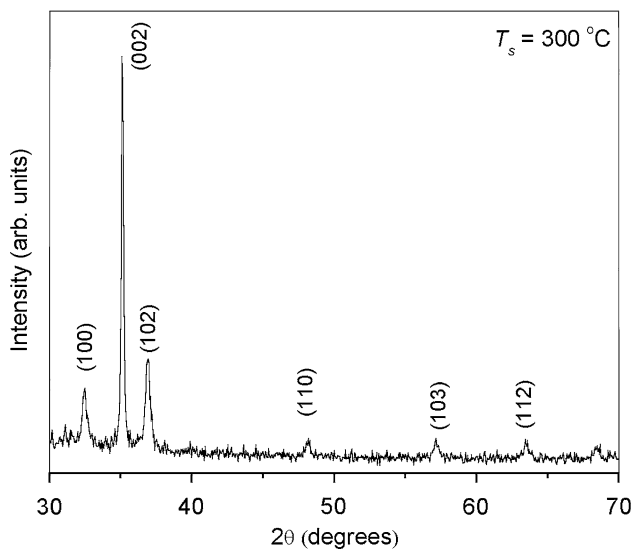
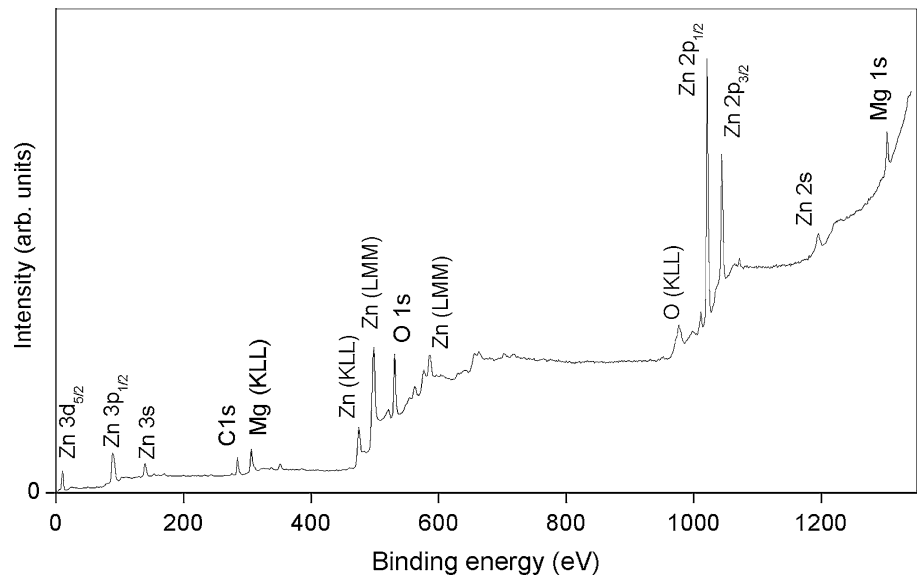


Fig. 2 XRD spectrum of $\text{Zn}_{0.76}\text{Mg}_{0.24}\text{O}$ film

grown at temperatures <300 °C might be due to the lower crystallinity leading to more light scattering [8, 9]. Further, the fundamental absorption edge was shifted towards shorter wavelengths with the increase of deposition temperature, suggesting the widening of energy band gap in the films due to the improvement in the crystalline quality.

The absorption coefficient, α in the strong absorption region was evaluated from the optical transmittance data using Lambert's principle,

$$\alpha = -(1/t) \ln T, \quad (1)$$

where T is the transmittance and ' t ' is the thickness of the film. The absorption coefficient was found to be $>10^5 \text{ cm}^{-1}$ above the absorption edge. Figure 5 shows the

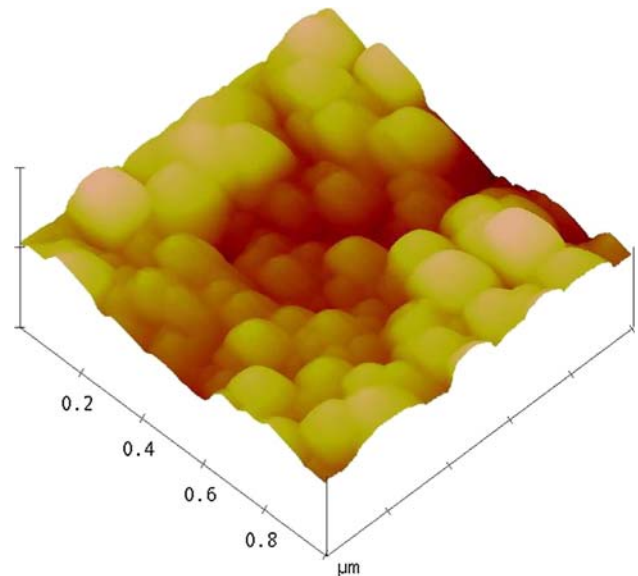


Fig. 3 AFM image of $\text{Zn}_{0.76}\text{Mg}_{0.24}\text{O}$ films grown at 300 °C

spectral dependence of absorption coefficient in $\text{Zn}_{0.76}\text{Mg}_{0.24}\text{O}$ films. It could be observed that the absorption coefficient increased with the increase of photon energy and showed a tail in the low energy region ($h\nu < E_g$). This indicated that there is a high concentration of impurity states causing the band structures to perturb, resulting in a prolonged tail extending into the energy gap in polycrystalline films [10]. An exponential increase in the absorption edge follows the Urbach empirical equation [11],

$$\alpha = \alpha_0 \exp(h\nu/E_0) \quad (2)$$

where E_0 and α_0 are characteristic parameters of the material, represented as the Urbach energy width of the exponential tail and Urbach absorption energy. According

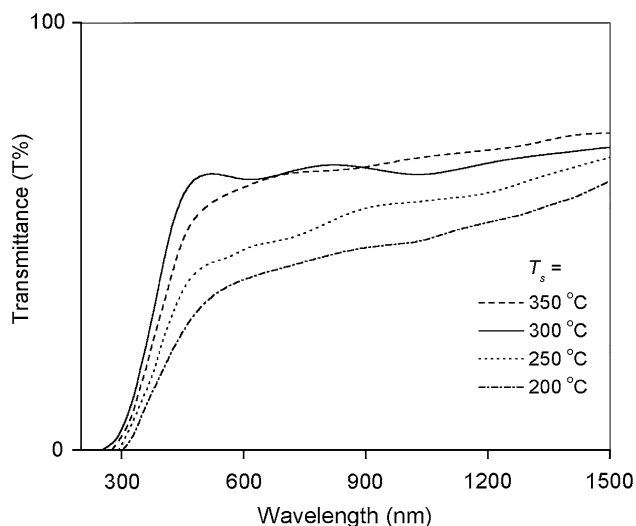


Fig. 4 Optical transmittance versus wavelength spectra of $\text{Zn}_{0.76}\text{Mg}_{0.24}\text{O}$ films deposited at different substrate temperatures

to the Eq. (3), the plot of $\ln \alpha$ as a function of photon energy, $h\nu$ gives a straight line. The Urbach tails were observed for all the films grown at temperatures from 200 to 350 °C. The values of E_0 decreased from 387 to 224 meV with the increase of substrate temperature from 200 to 300 °C and it increased to 249 meV with further increase of substrate temperature to 350 °C. Melsheimer et al. [10] interpreted the width, E_0 as a function of structural disorder, which decreases with the increase of structural order. This behaviour of E_0 is in good agreement with our present observation for all the films.

The type of optical transition and the energy band gap of the films can be determined from the dependence of absorption coefficient, α on the photon energy, $h\nu$. The

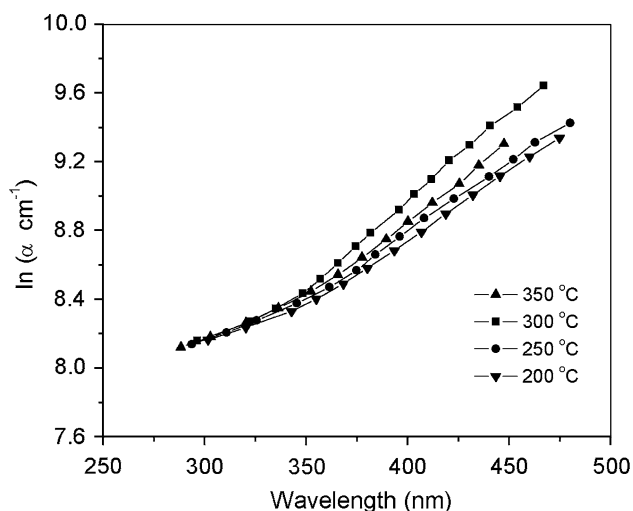


Fig. 5 The spectral dependence of absorption coefficient of $\text{Zn}_{0.76}\text{Mg}_{0.24}\text{O}$ films grown at different substrate temperatures

direct or indirect nature of optical transition between parabolic bands can be studied using the relation [11],

$$\alpha h\nu = A(h\nu - E_g)^x \quad (3)$$

where A is a constant, E_g is the optical energy band gap, $x = 1/2$ and $3/2$ for the direct allowed and direct forbidden transitions respectively. Further, the value of $x = 2$ and 3 for the indirect allowed and indirect forbidden transitions respectively. In the present study, the variation of absorption coefficient with photon energy followed the relation where $x = 1/2$ indicating that the transition corresponds to a directly allowed electronic transition. Figure 6 shows a plot of $(\alpha h\nu)^2$ versus $h\nu$ for the films deposited at 300 °C and the variation of energy band gap in the films deposited at three different temperatures is shown in the inset. The energy band gap was determined by extrapolation of the linear portion of $(\alpha h\nu)^2$ versus $h\nu$ plot onto the energy axis. An increase of band gap from 3.51 to 3.87 eV with the increase of deposition temperature was observed.

The refract index of the layers, determined from the spectroscopic measurements, increased with the increase of substrate temperature and it varied in the range, 1.81–1.98, as shown in Fig. 7. This might be due to the improvement in the packing density of the grains with the increase of substrate temperature. These values are in close agreement with the data reported by Chen et al. [12] for reactive electron beam evaporated films, but less than the data reported for the magnetron sputtered films [13]. The extinction coefficient (k) was directly calculated from the absorption coefficient by using the relation [14],

$$k = \frac{\alpha \lambda}{4\pi} \quad (4)$$

The variation of extinction coefficient with substrate temperature is shown in Fig. 7. The extinction coefficient

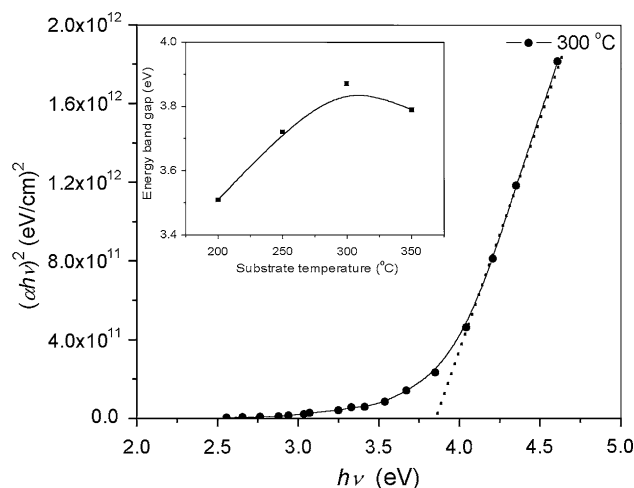


Fig. 6 Plot of $(\alpha h\nu)^2$ versus $h\nu$ (inset: variation of energy band gap with substrate temperature)

varied in the range, 10^{-1} – 10^{-2} with the increase of substrate temperature in the range, 200–300 °C. Such decrease of k with substrate temperature implies weaker photon absorption by electron transitions in the band gap due to the decrease of intermediate defect levels. The higher value of extinction coefficient for the films deposited at lower temperatures (<300 °C) may be due to the crystallographic defects such as grain boundaries and voids present in the layers that appear to reduce with the increase of deposition temperature. This might be due to an improvement in the crystallinity of the films with the increase of deposition temperature leading to minimum imperfections.

The room temperature photoluminescence spectra of $\text{Zn}_{0.76}\text{Mg}_{0.24}\text{O}$ films recorded in the wavelength range, 350–600 nm when excited using a light of wavelength 335 nm is shown in Fig. 8. The films showed an intense UV peak at 373 nm along with a broad luminescence in the visible region. Also, the intensity of UV luminescence increased with deposition temperature along with a blue shift in both the luminescence peaks. Generally, the PL spectra of ZnO consisted of an intense UV emission band and a broad visible emission band. The UV emission is attributed to the exciton recombination whereas the visible emission is mainly due to the structural defects that are related to the deep-levels associated with zinc and oxygen vacancies and the corresponding interstitials [15]. The observed blue shift in the UV band could be due to the increase of band gap with the increase of deposition temperature of $\text{Zn}_{0.76}\text{Mg}_{0.24}\text{O}$ films. It is well known that the light emission intensity was determined by the radiative and nonradiative transitions [16]. In the as-grown films, there are many lattice defects and surface defects that can produce various nonradiative centres and reduce the light emission from the material. Therefore, the films deposited at 200 °C showed a lower luminescence intensity in the UV region. The intensity of this emission enhanced

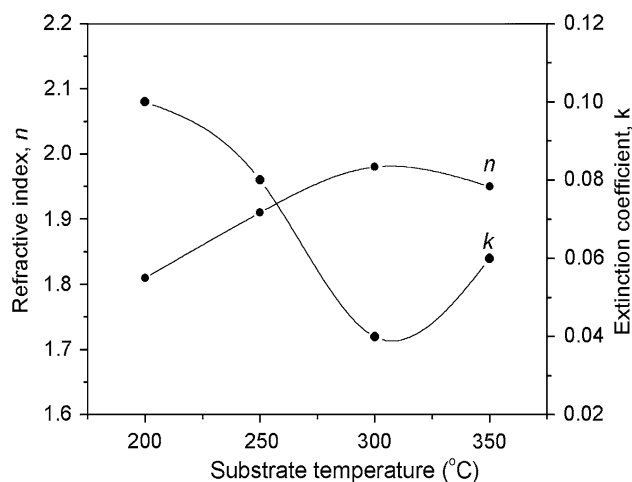


Fig. 7 Variation of n and k of $\text{Zn}_{0.76}\text{Mg}_{0.24}\text{O}$ films with T_s

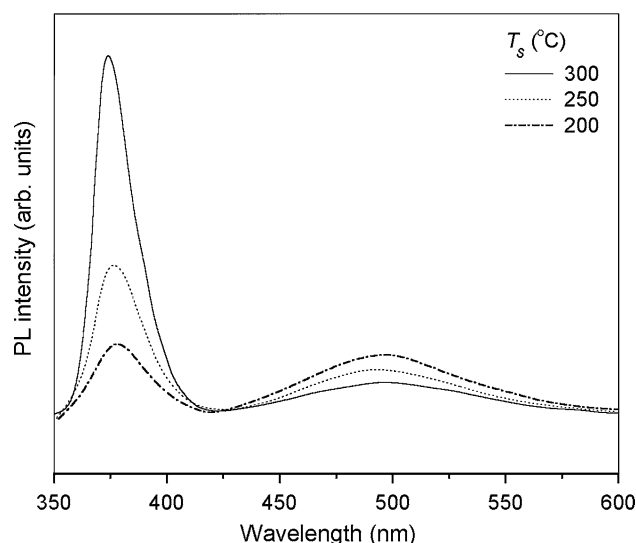


Fig. 8 Photoluminescence spectra of $\text{Zn}_{0.76}\text{Mg}_{0.24}\text{O}$ films grown at different substrate temperatures

remarkably with the increase of substrate temperature to 300 °C along with a decrease of the intensity of the green band, which implies that the raise of T_s could reduce the traps for nonradiative transition resulting from the improvement in crystallinity.

All the as-grown $\text{Zn}_{0.76}\text{Mg}_{0.24}\text{O}$ films showed a high resistivity that varied in the range, 10^6 – 10^5 Ωcm when the substrate temperature increases from 200 to 300 °C. This might be due to the decrease of native defects that act as trapping centres at lower growth temperatures. Further, temperature dependent conductivity measurements were made in the temperature range, 300–450 K in order to evaluate the activation energy of the charge carriers. Figure 9 shows the variation of $\ln\sigma$ versus inverse of T for a representative film prepared at 300 °C. The activation energy evaluated from the slopes of the plot of $\ln\sigma$ versus T^{-1} decreased from 0.714 to 0.403 eV with the increase of substrate temperature from 200 to 300 °C. This could be due to the decrease of crystallographic defects and improvement in the crystallinity of the grown layers. The activation energy values calculated in this study were lower than the reported data for the films deposited by metal organic chemical vapour deposition [17]. This might be due to the better crystallinity of the films grown in the present study.

4 Conclusions

Polycrystalline $\text{Zn}_{0.76}\text{Mg}_{0.24}\text{O}$ thin films have been prepared successfully by spray pyrolysis method. The compositional analysis made by XPS measurements indicated that the stoichiometric Mg doping percentage was achieved

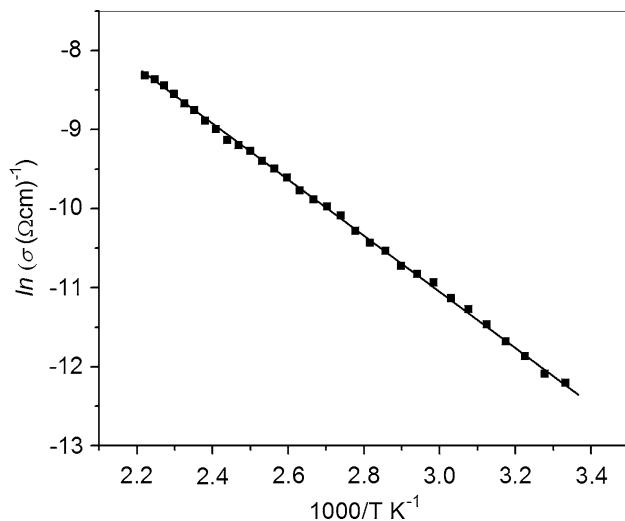


Fig. 9 The variation of $\ln\sigma$ versus T^{-1} for the films prepared at 300 °C

at a substrate temperature of 300 °C. All the films exhibited hexagonal wurtzite structure with (002) orientation and its intensity was maximum at the substrate temperature of 300 °C that decreased thereafter. The surface morphological analysis revealed that the films prepared at 300 °C had good crystallinity and uniform distribution of the grains with a lower surface roughness of 6.2 nm. The optical analysis revealed that the substrate temperature had a significant influence on the optical transmittance of $\text{Zn}_{0.76}\text{Mg}_{0.24}\text{O}$ layers, which showed the highest transmittance of >60% with an optical band gap of 3.87 eV. The layers exhibited a broad visible band with a sharp UV band and a small blue shift in the luminescence peak was observed with the increase of substrate temperature.

Finally, the as-grown $\text{Zn}_{0.76}\text{Mg}_{0.24}\text{O}$ layers showed a lower electrical resistivity of $10^5 \Omega\text{cm}$ with a lower activation energy of 0.403 eV.

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